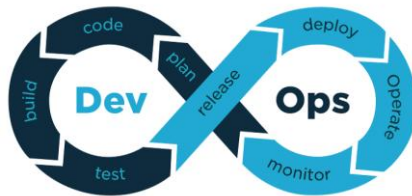


FATIMA JINNAH WOMEN UNIVERSITY
RAWALPINDI



Cloud DevOps



Lab 01

Title: GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Name: Rabbia Noor

Registration No.: 2023-BCS-70

Course Code: CS-649

GitHub Username: Rabbia-Noor

Section: A

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Getting Started: Folder Structure

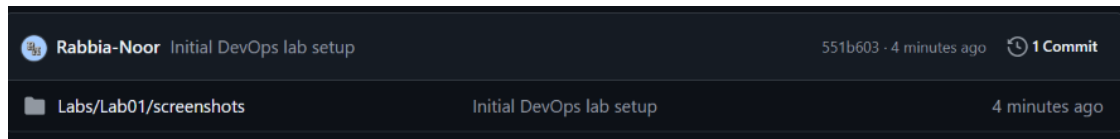


Figure 1: Directory structure `Labs/Lab01/screenshots` created inside the DevOps repository (`lab01_folder_structure.png`).

Task 1: Create and Configure Your GitHub Account



Figure 2: GitHub profile showing the profile name and username Rabbia-Noor (`github_profile.png`).

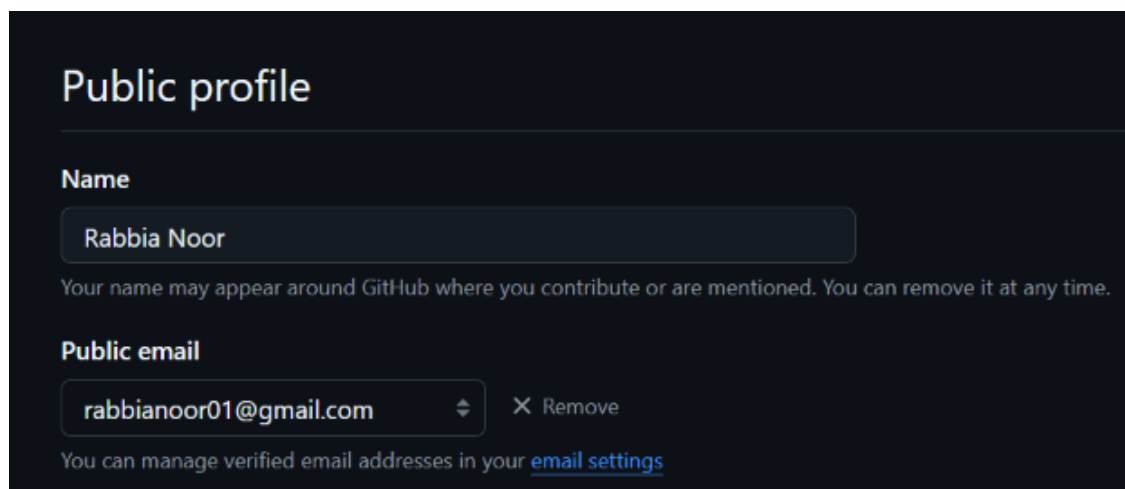


Figure 3: GitHub Settings > Public profile showing the complete actual name in the Name field (`github_actual_name.png`).

Task 2: Register in the Student Marks Portal

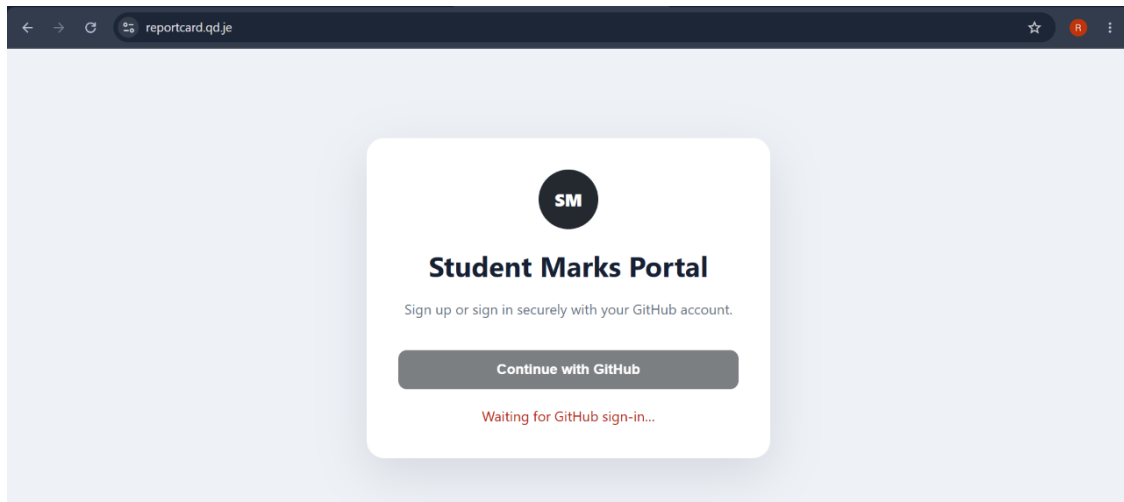


Figure 4: Student Marks Portal login page with the Continue with GitHub option (portal_login_page.png).

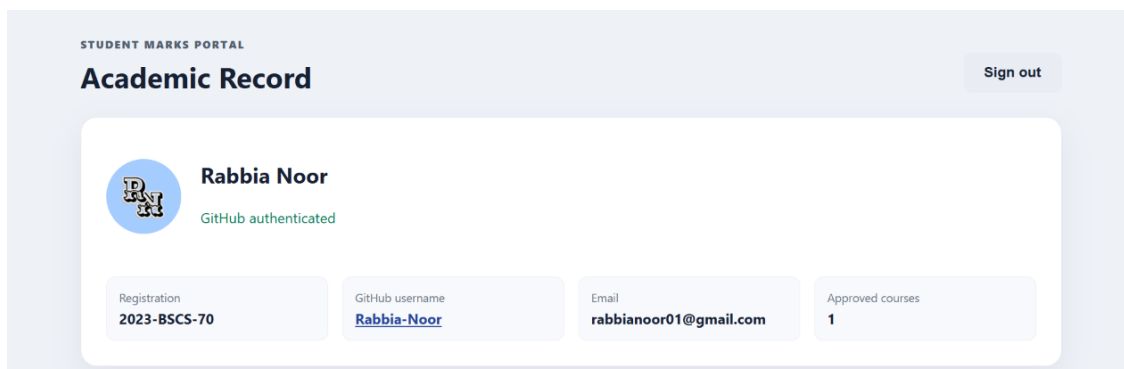


Figure 5: GitHub profile information, registration number and username shown inside the portal (portal_github_profile.png).

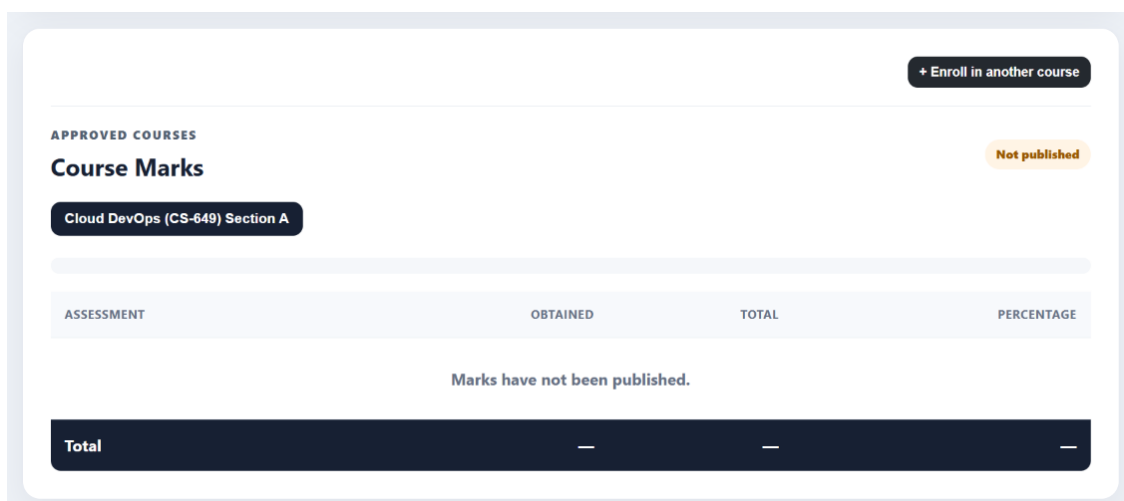


Figure 6: Submitted enrollment for Cloud DevOps, Section A, shown in the portal (portal_enrollment_submitted.png).

Task 3: Download Virtualization Software and Ubuntu Server

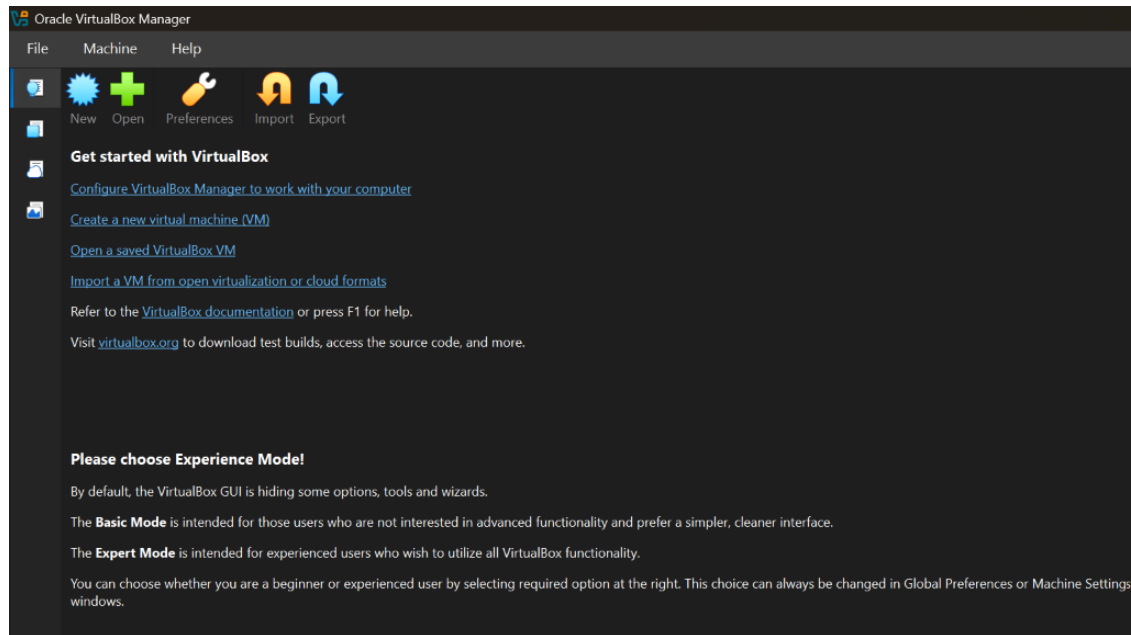


Figure 7: Virtualization software (Oracle VirtualBox Manager) installed and running (vmware_workstation.png).

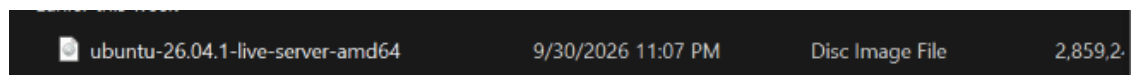


Figure 8: Downloaded Ubuntu Server LTS ISO file: ubuntu-26.04.1-live-server-amd64 (ubuntu_server_iso.png).

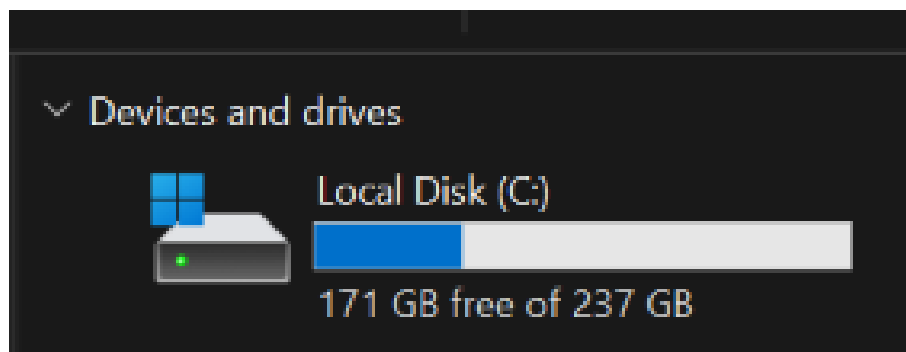


Figure 9: Local Disk (C:) showing 171 GB free of 237 GB, more than the required 20 GB (available_storage.png).

Task 4: Create the Ubuntu Server Virtual Machine

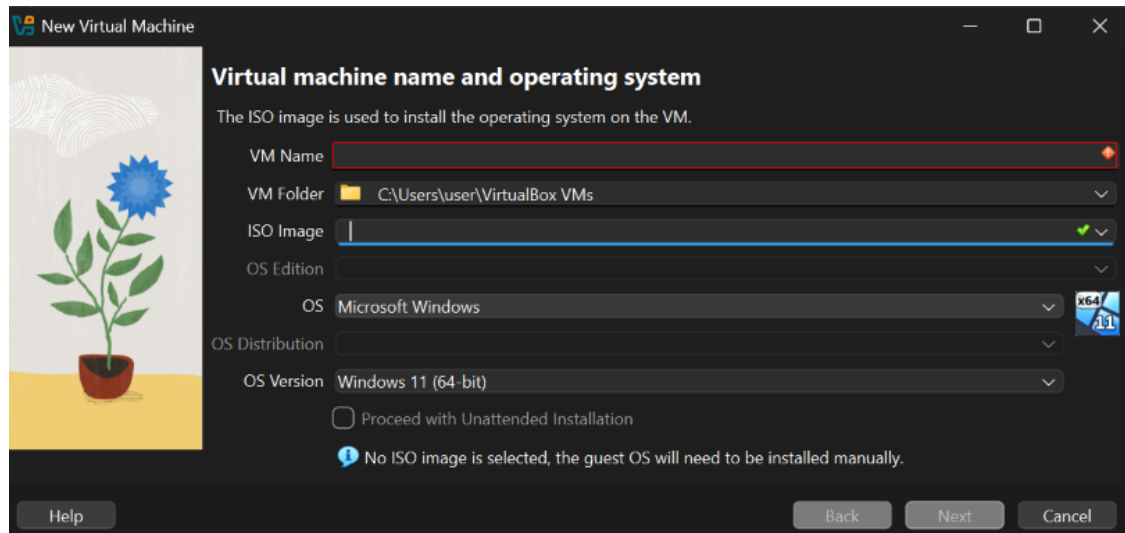


Figure 10: New Virtual Machine wizard opened (vm_creation_wizard.png).

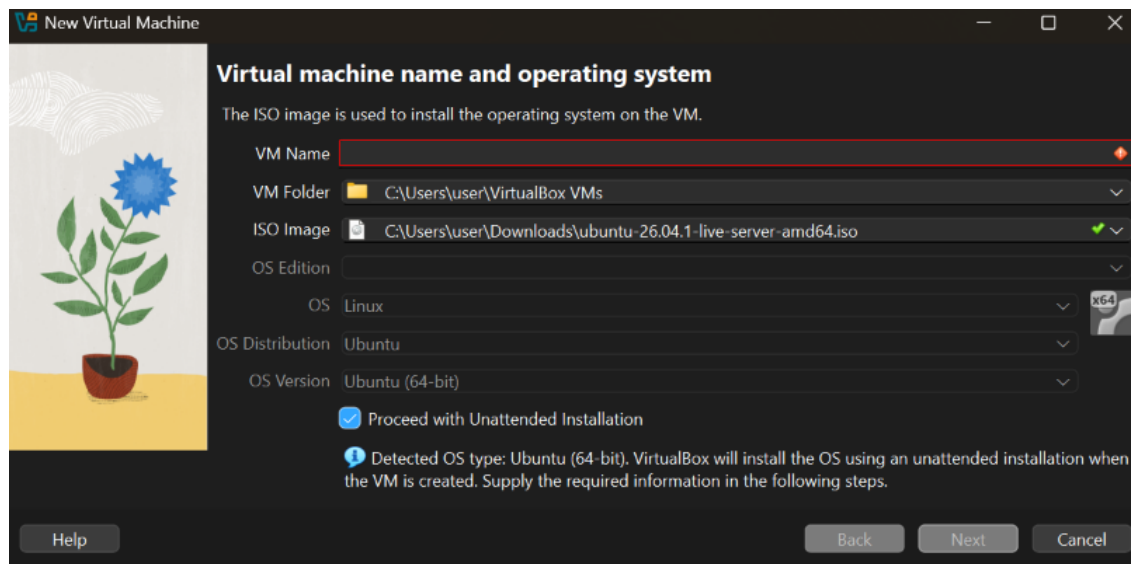


Figure 11: Virtual machine name and operating system configuration selected (vm_typical_configuration.png).

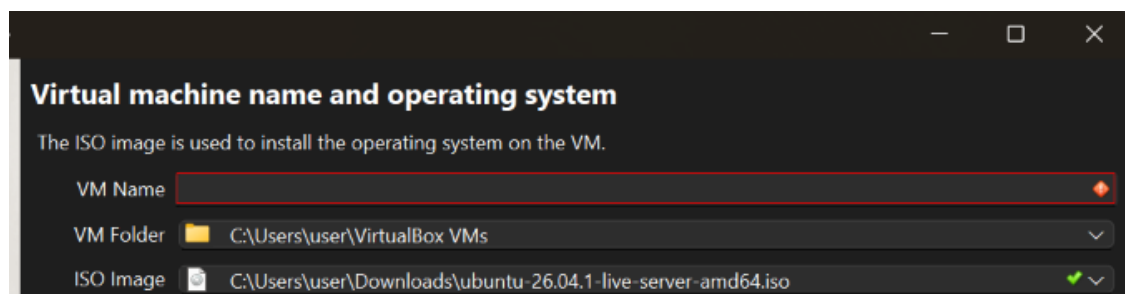


Figure 12: Ubuntu Server ISO path selected in the ISO Image field (vm_ubuntu_iso_selected.png).

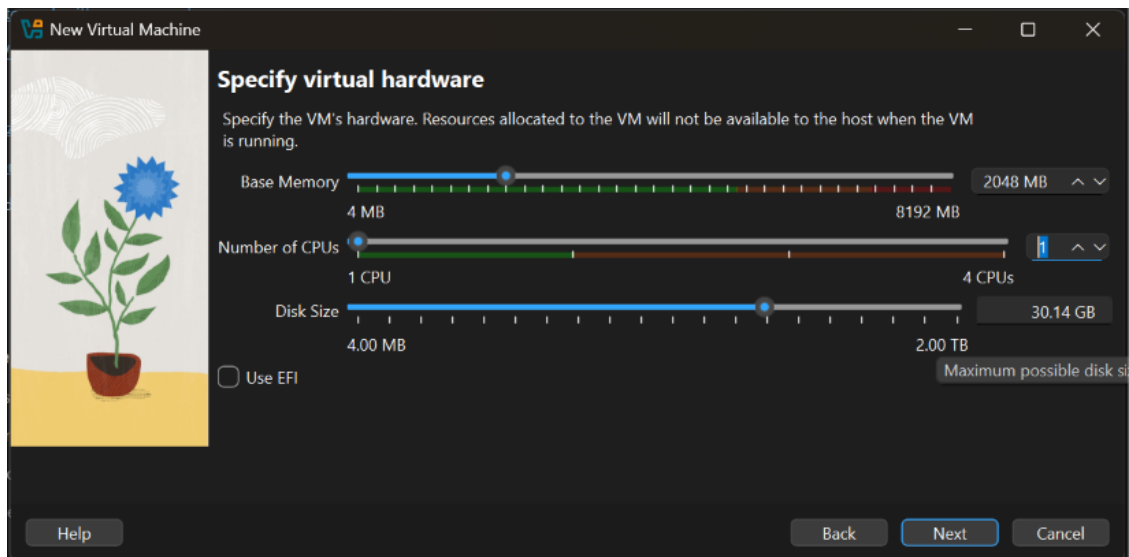


Figure 13: Virtual hardware and virtual disk size (more than 20 GB) allocated for the VM.

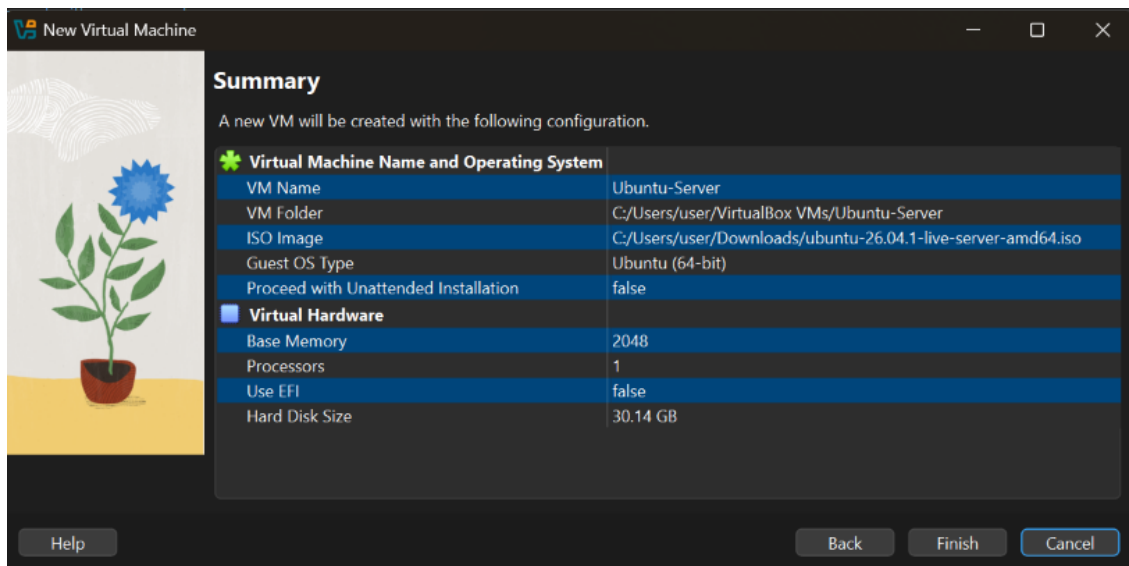


Figure 14: Final virtual machine configuration summary (vm_configuration_summary.png).

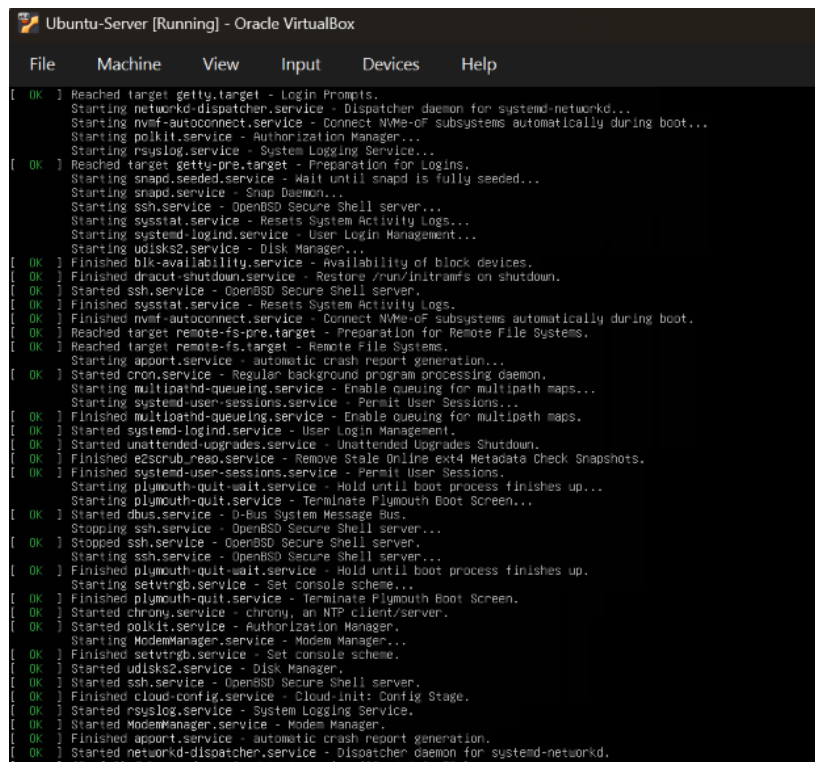


Figure 15: Ubuntu Server installer starting after the virtual machine was powered on (ubuntu_installer_boot.png).

Task 5: Choose the Installation Language

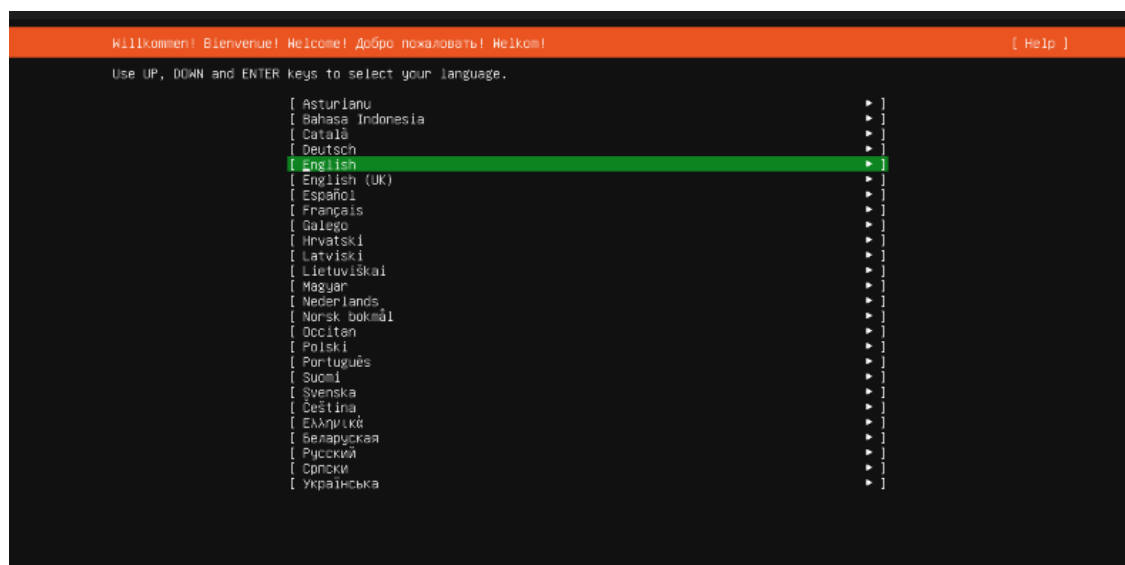


Figure 16: English selected as the installation language (ubuntu_language.png).

Task 6: Configure the Keyboard Layout

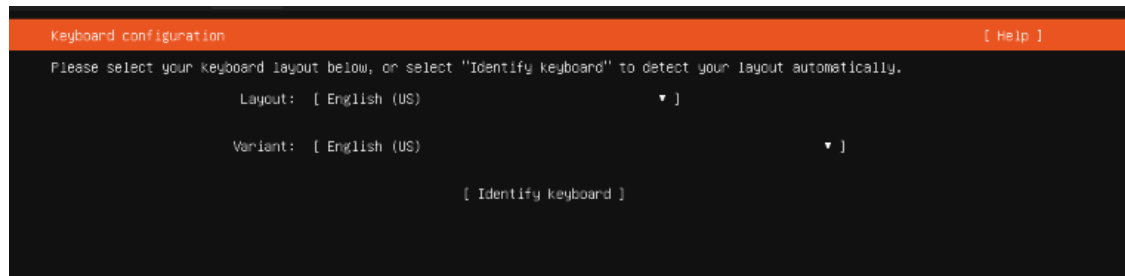


Figure 17: Keyboard layout set to English (US) (ubuntu_keyboard_layout.png).

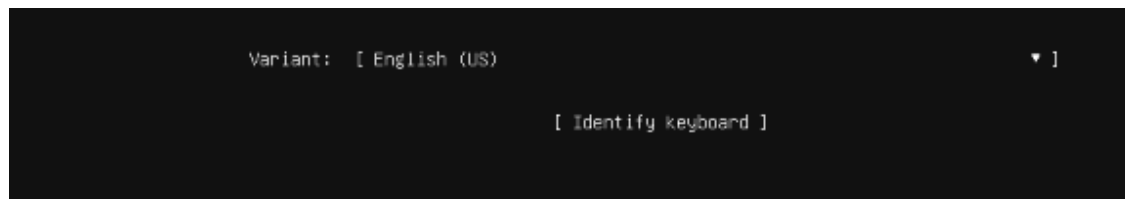


Figure 18: Keyboard variant set to English (US) (ubuntu_keyboard_variant.png).

Task 7: Choose the Ubuntu Server Installation Type

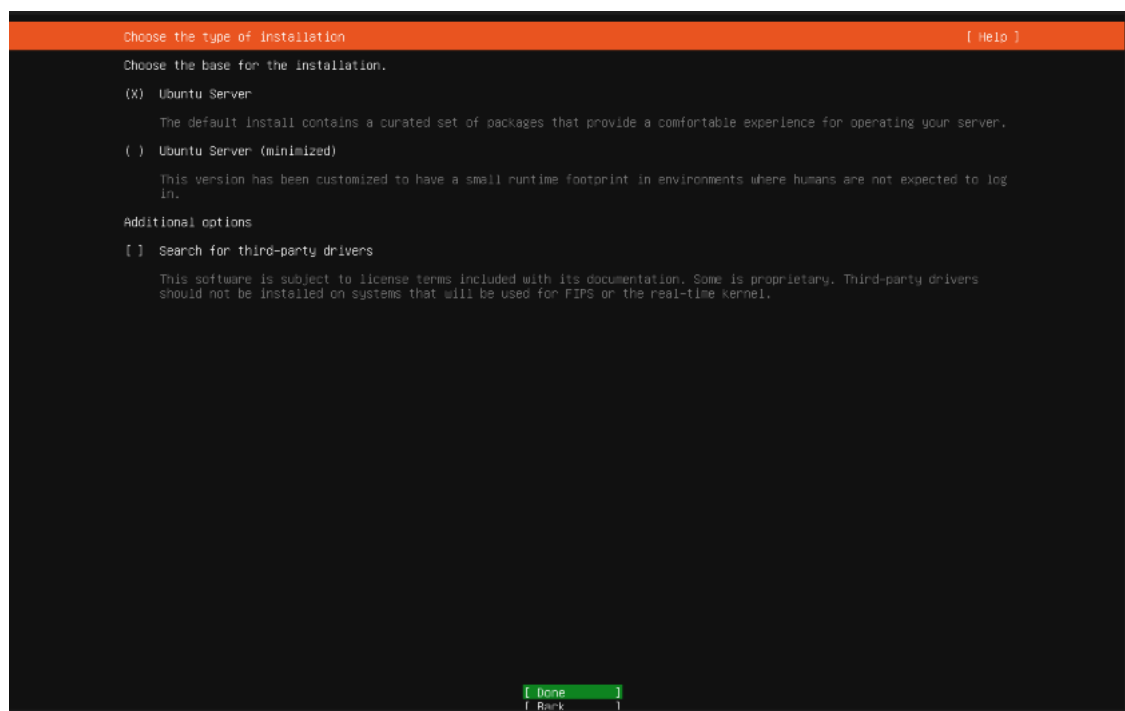


Figure 19: Normal Ubuntu Server installation type selected (ubuntu_installation_type.png).

Task 8: Configure the Network

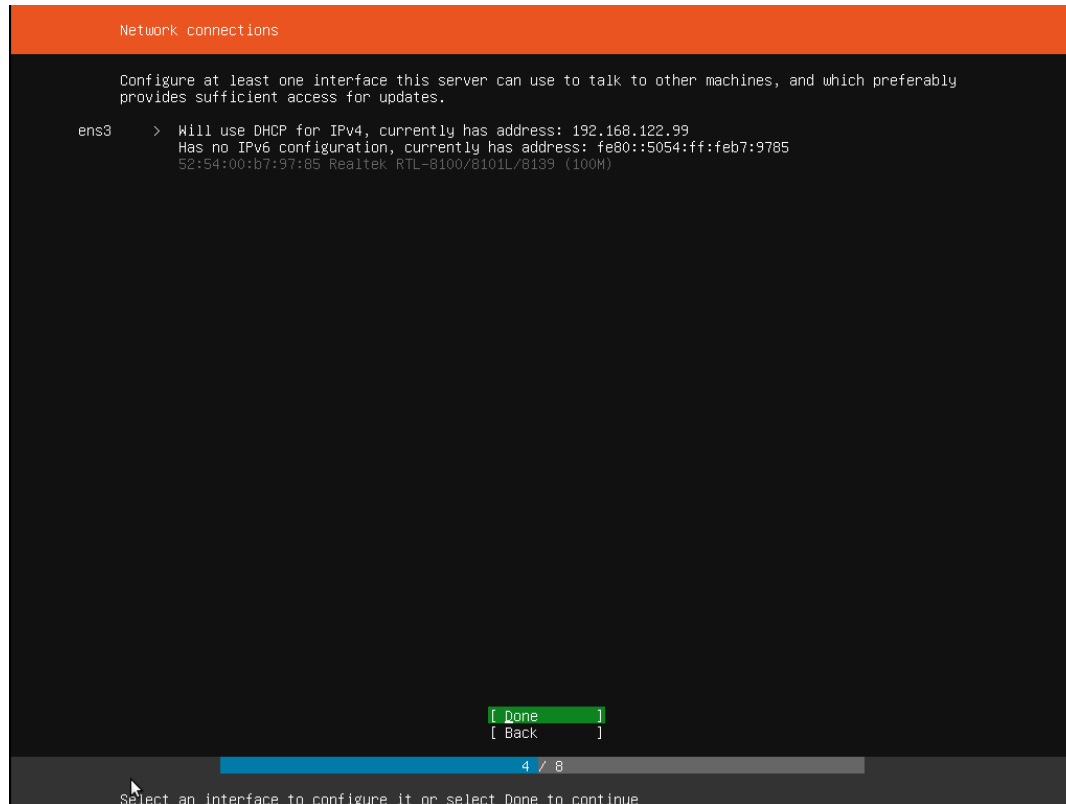


Figure 20: Detected network interface in the Ubuntu installer (*ubuntu_network_interface.png*).

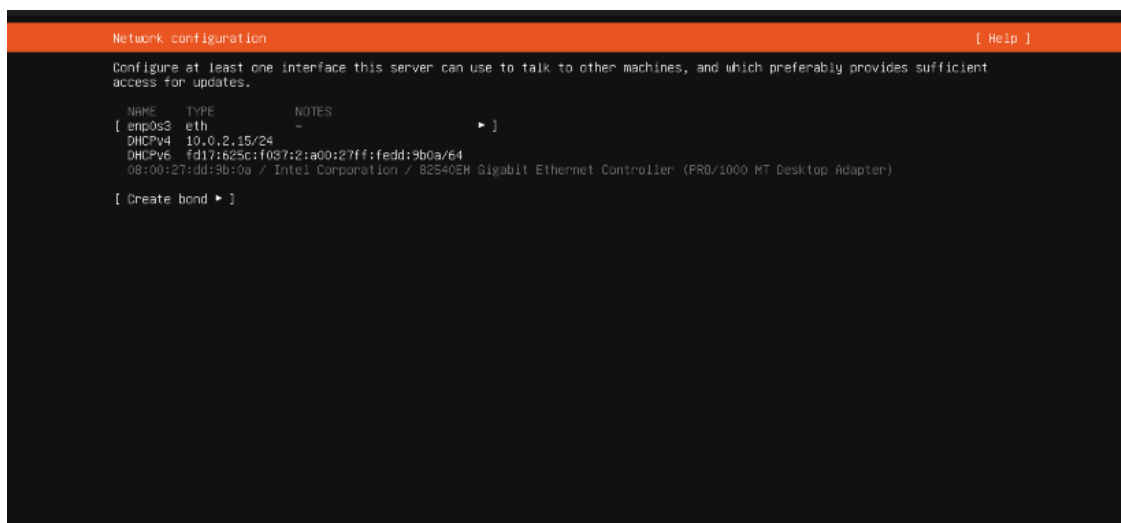


Figure 21: Network interface with an address assigned automatically through DHCP (*ubuntu_installer_network.png*).

Task 9: Configure Storage

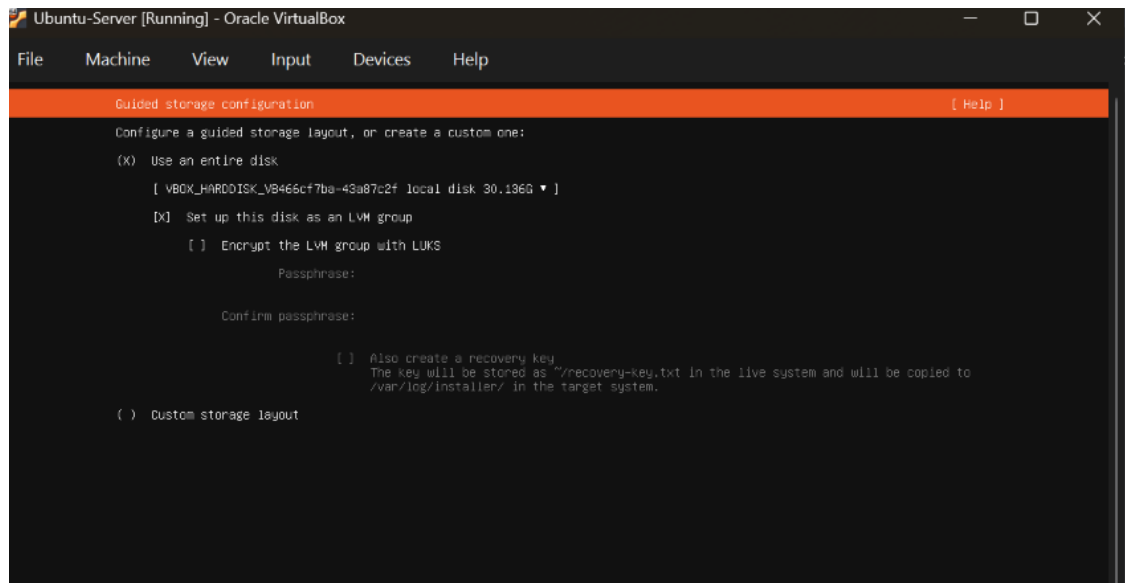


Figure 22: Guided storage option: use an entire disk (ubuntu_guided_storage.png).

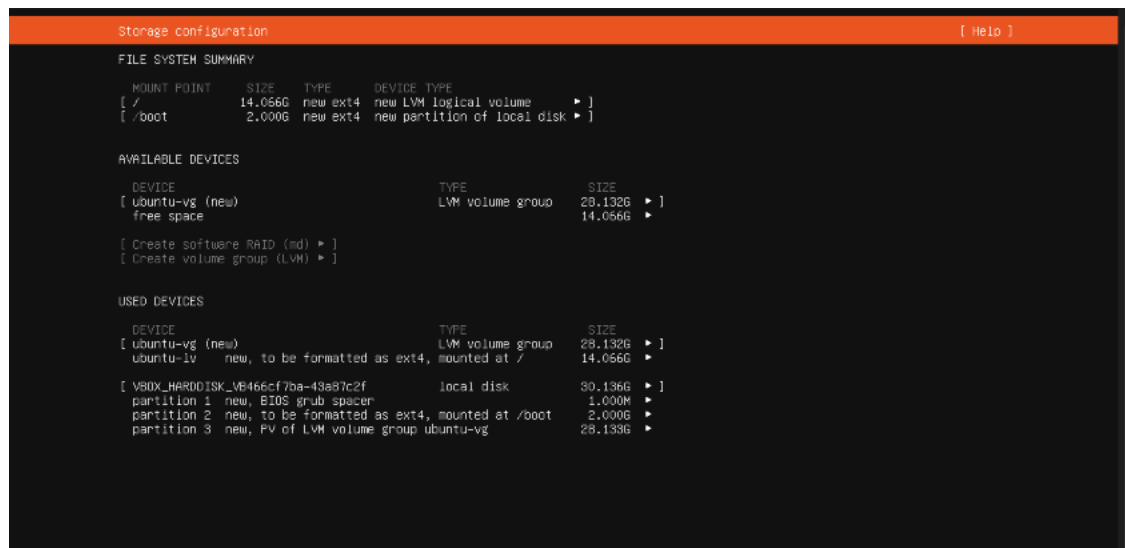


Figure 23: Storage configuration reviewed before continuing (ubuntu_storage_configuration.png).

Task 10: Select the Installation Disk



Figure 24: Virtual disk selected for the installation (ubuntu_selected_disk.png).

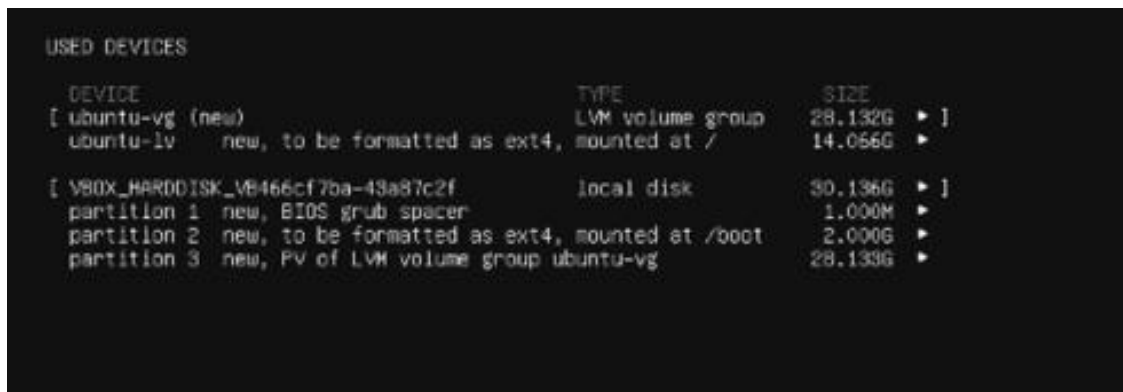


Figure 25: Disk name and size confirming the correct virtual disk (ubuntu_disk_details.png).

Task 11: Review the Partitions

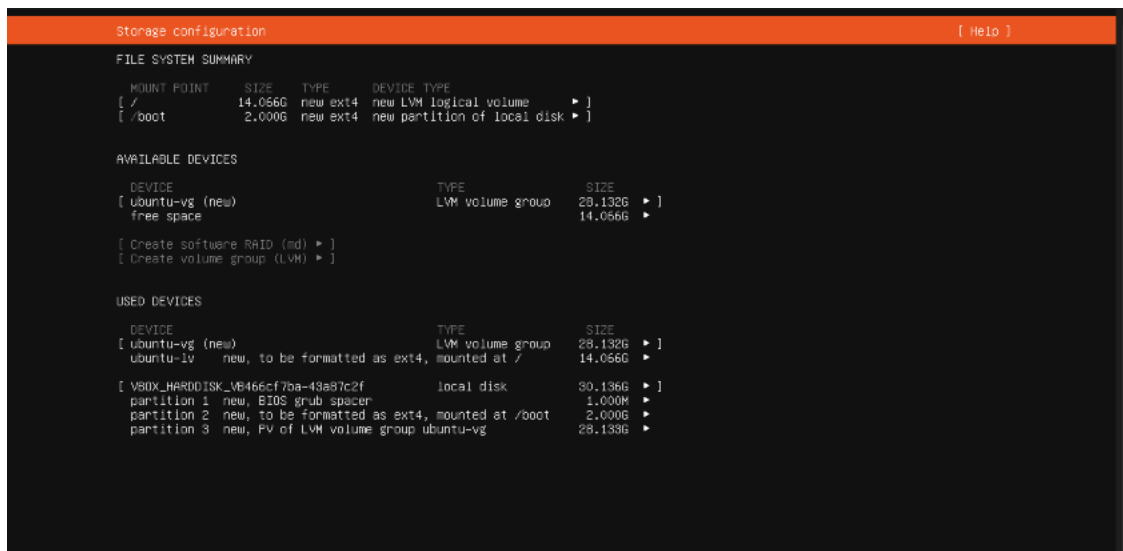


Figure 26: Proposed partition layout on the virtual disk (ubuntu_partition_layout.png).

Task 12: Confirm the Storage Changes

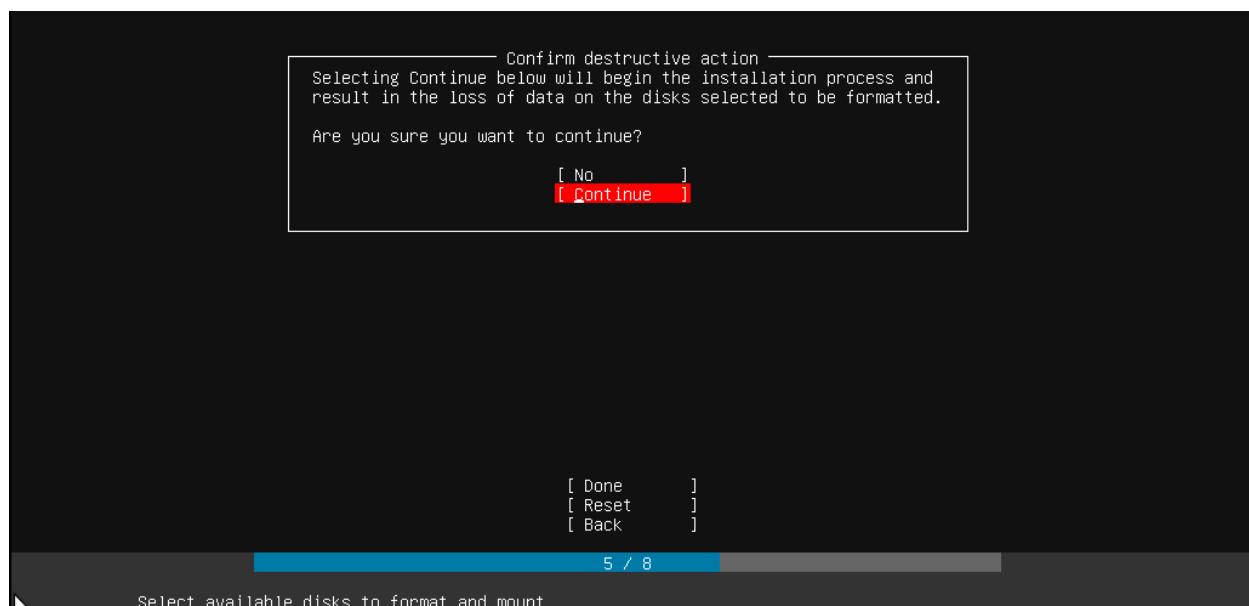


Figure 27: Final destructive-action warning before formatting the disk (ubuntu_storage_warning.png).

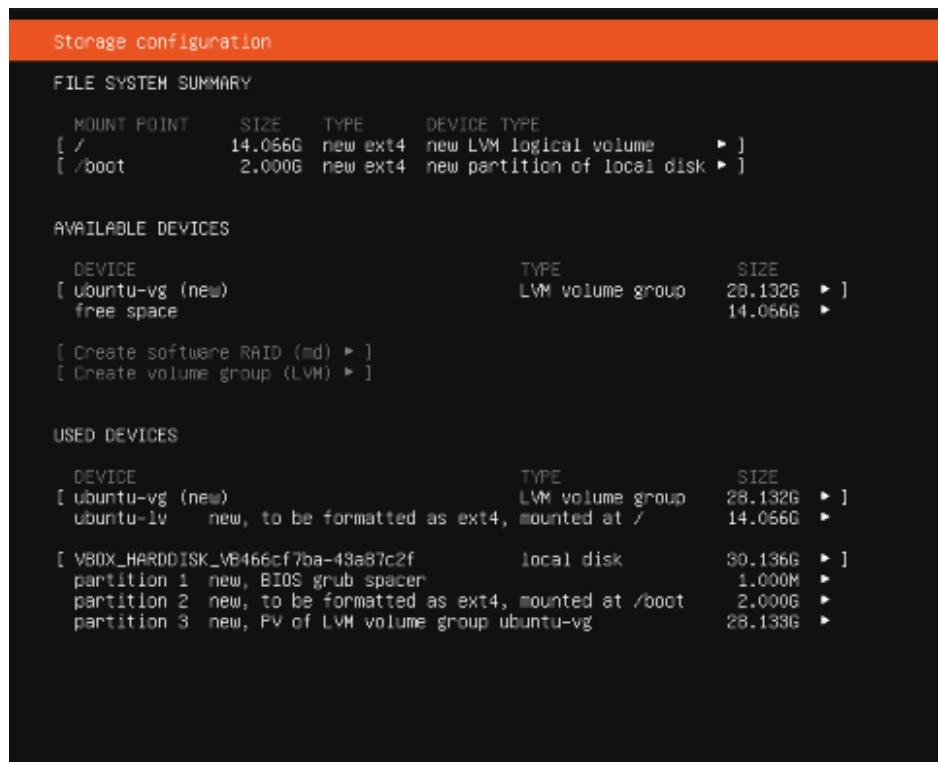


Figure 28: Storage changes reviewed and confirmed on the virtual disk (ubuntu_storage_confirmed.png).

Task 13: Configure the Ubuntu Profile

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:

Pick a username:

Choose a password:

Confirm your password:

[Done]

Figure 29: Complete actual name entered in the Your name field (ubuntu_actual_name.png).

The screenshot shows a terminal window titled "Profile configuration" with a "[Help]" link in the top right. The instructions state: "Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo." The form contains the following fields and values:

- Your name:** Rabbia Noor
- Your servers name:** ubuntu (with a subtext: "The name it uses when it talks to other computers.")
- Pick a username:** (empty field, with a red error message "Username missing" below it)
- Choose a password:** (empty field, with a red error message "Password must be set" below it)
- Confirm your password:** (empty field)

Figure 30: Server name set to ubuntu (ubuntu_server_name.png).

This screenshot shows the same "Profile configuration" terminal window. The "Pick a username" field is now filled with "rabbia-noor". The "Choose a password" field remains empty with the "Password must be set" error message. At the bottom of the terminal window, a "[Done]" link is visible.

Figure 31: Ubuntu username entered, matching the GitHub username (ubuntu_username.png).

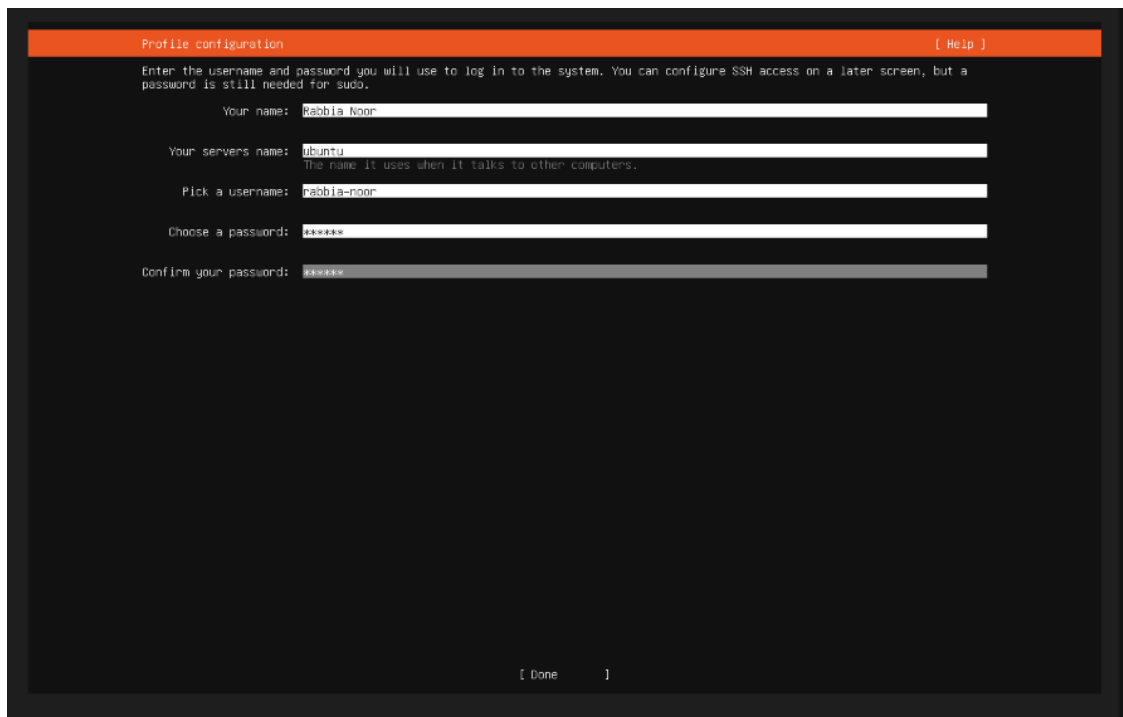


Figure 32: Completed profile configuration page (ubuntu_profile_setup.png).

Task 14: Complete the Software Installation

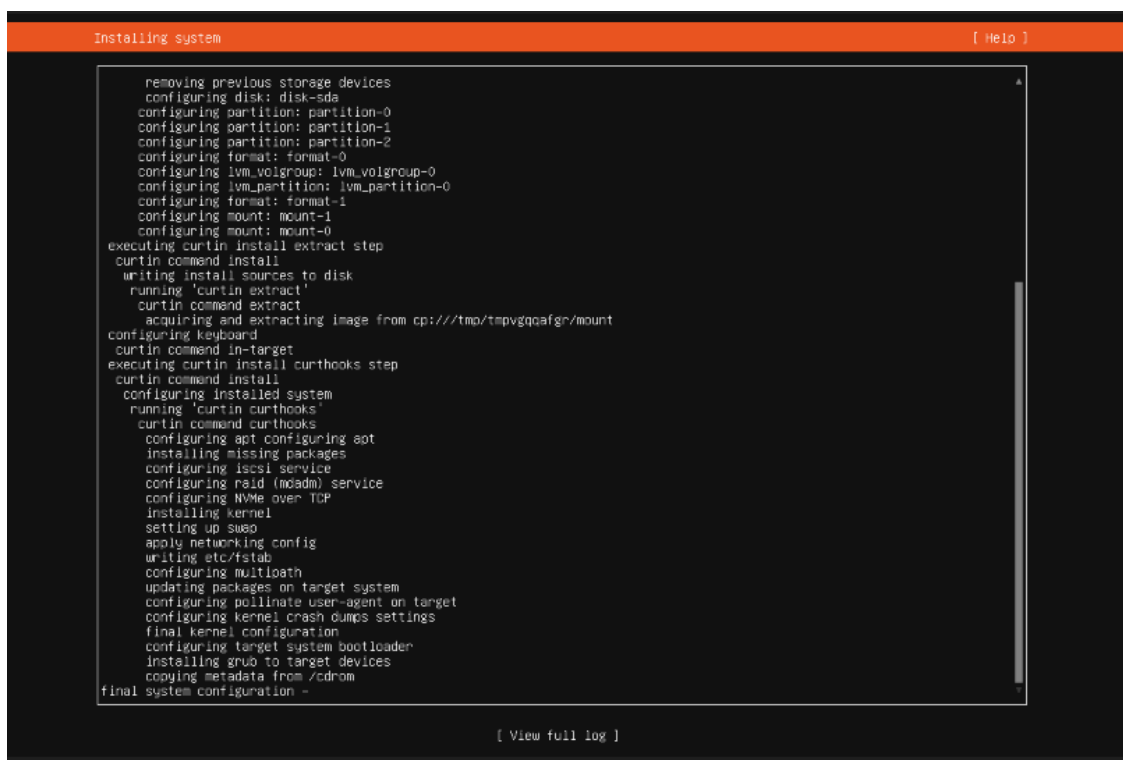


Figure 33: Ubuntu Server installation in progress (ubuntu_installation_progress.png).

Task 15: Reboot and Verify the Ubuntu Identity

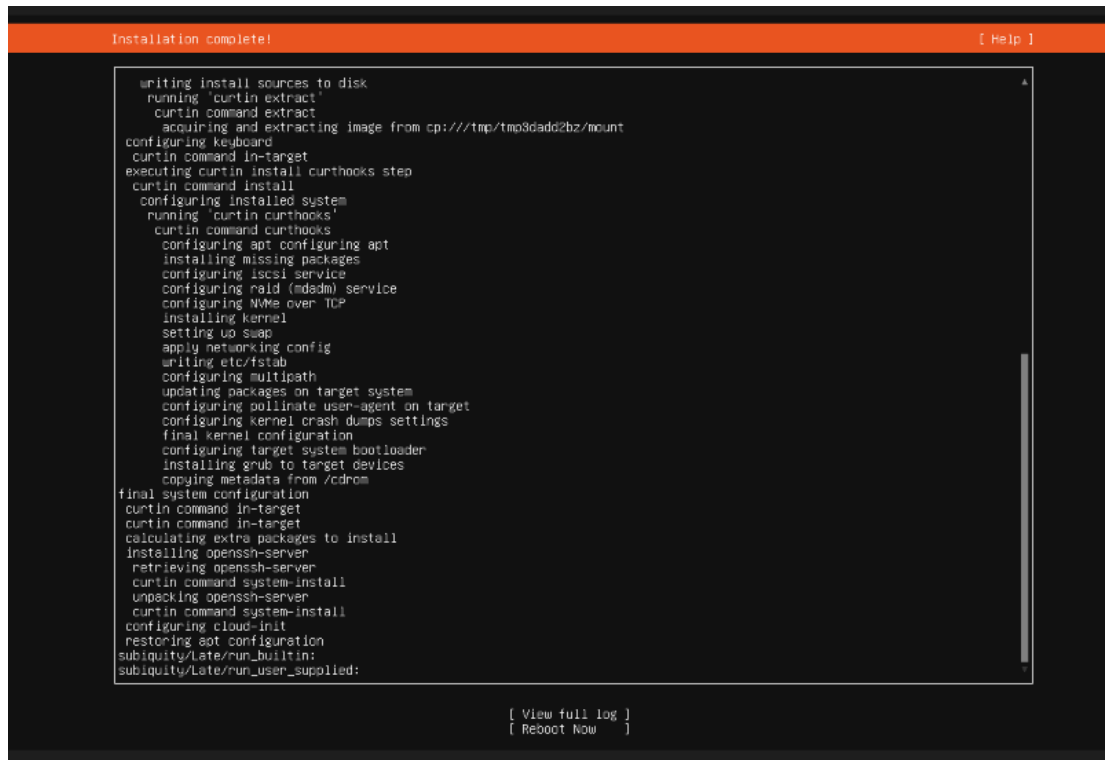


Figure 34: Installation complete screen with the Reboot Now option (ubuntu_installation_complete.png).

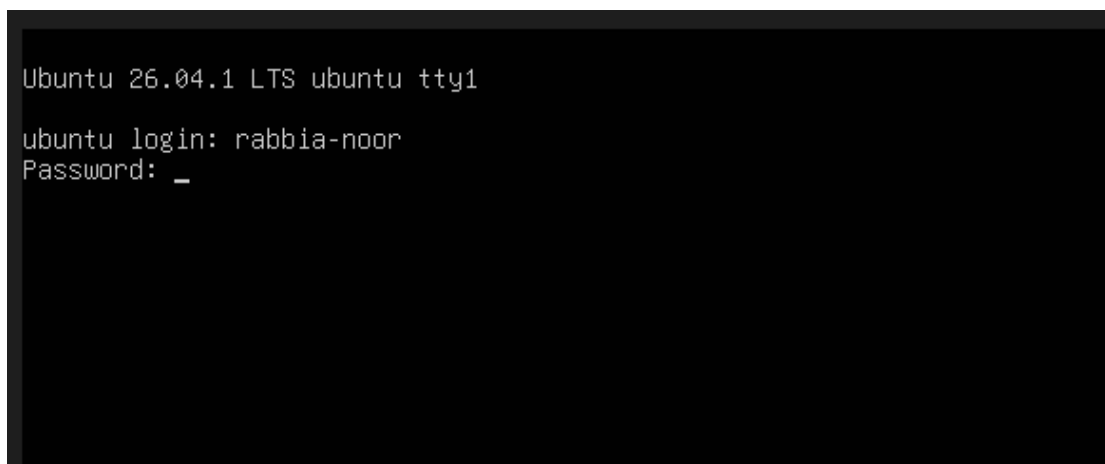


Figure 35: Ubuntu login screen after reboot (ubuntu_login_screen.png).

```
Ubuntu 26.04.1 LTS ubuntu tty1
ubuntu login: rabbia-noor
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Aug 26 04:51:42 UTC 2026

System load:  1.07      Processes:      26
Usage of /home: unknown  Users logged in:  0
Memory usage:  0%      IPv4 address for eth0: 10.10.10.2
Swap usage:    0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

rabbia-noor@ubuntu:~$
```

Figure 36: Successful terminal login to Ubuntu Server (ubuntu_terminal_login.png).

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

rabbia-noor@ubuntu:~$
```

Figure 37: Terminal prompt showing rabbia-noor@ubuntu, verifying the identity (ubuntu_identity_verified.png).

Task 16: Find the Ubuntu Server IP Address

```
rabbia-noor@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:dd:9b:0a brd ff:ff:ff:ff:ff:ff
    altname enx080027dd9b0a
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85910sec preferred_lft 85910sec
    inet6 fd17:625c:f037:2:a00:27ff:fedd:9b0a/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86297sec preferred_lft 14297sec
    inet6 fe80:a00:27ff:fedd:9b0a/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
rabbia-noor@ubuntu:~$ s_
```

Figure 38: The ip addr command and its output run inside Ubuntu Server (ubuntu_ip_addr_command.png).

```

    valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:dd:9b:0a brd ff:ff:ff:ff:ff:ff
    altname enx080027dd9b0a
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85910sec preferred_lft 85910sec
    inet6 fd17:625c:f037:2:a00:27ff:fedd:9b0a/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86297sec preferred_lft 14297sec
    inet6 fe80::a00:27ff:fedd:9b0a/64 scope link proto kernel_l1
    valid_lft forever preferred_lft forever
rabbia-noor@ubuntu:~$ s

```

Figure 39: IPv4 address of the active network interface shown after `inet` (`ubuntu_ip_address.png`).

Task 17: Connect to Ubuntu Server from Windows

```

rabbia-noor@ubuntu: ~
C:\Users\user>ssh rabbia-noor@192.168.0.104
The authenticity of host '192.168.0.104 (192.168.0.104)' can't be established.
ED25519 key fingerprint is SHA256:wssnTY1ZewW97Z34KKKeDBK82e0gVUa3zyRPBjXcgoAQ.

```

Figure 40: SSH command run from Windows Command Prompt (`windows_ssh_command.png`).

```

rabbia-noor@ubuntu: ~
C:\Users\user>ssh rabbia-noor@192.168.0.104
The authenticity of host '192.168.0.104 (192.168.0.104)' can't be established.
ED25519 key fingerprint is SHA256:wssnTY1ZewW97Z34KKKeDBK82e0gVUa3zyRPBjXcgoAQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.104' (ED25519) to the list of known hosts.
rabbia-noor@192.168.0.104's password:
Permission denied, please try again.
rabbia-noor@192.168.0.104's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sat Oct  3 08:49:51 AM UTC 2026

System load:  0.0           Processes:    105
Usage of /:   35.5% of 13.74GB Users logged in: 0
Memory usage: 13%          IPv4 address for enp0s3: 192.168.0.104
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

```

Figure 41: Server fingerprint accepted by typing yes (`windows_ssh_fingerprint.png`).

```
Permission denied, please try again.
rabbia-noor@192.168.0.104's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Sat Oct  3 08:49:51 AM UTC 2026

System load:  0.0      Processes:            105
Usage of /:   35.5% of 13.74GB Users logged in:       0
Memory usage: 13%      IPv4 address for enp0s3: 192.168.0.104
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sat Oct  3 08:04:41 2026 from 10.0.2.15
rabbia-noor@ubuntu:~$ |
```

Figure 42: Successful remote SSH login to Ubuntu Server (windows_ssh_login.png).

```
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Sat Oct  3 08:49:51 AM UTC 2026

System load:  0.0      Processes:            105
Usage of /:   35.5% of 13.74GB Users logged in:       0
Memory usage: 13%      IPv4 address for enp0s3: 192.168.0.104
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sat Oct  3 08:04:41 2026 from 10.0.2.15
```

Figure 43: Remote prompt confirming the identity rabbia-noor@ubuntu (windows_ssh_identity.png).